Lesson 4.1: Data preprocessing and model training (Pandas, NumPy, Scikit-Learn, PyTorch)

What you learn

- **Preprocessing.** Clean missing values, fix types, encode categories and scale numbers where the model needs it.
- **Train and test split.** Train on one part of the data and measure on rows the model has not seen.
- **Scikit-Learn.** Classical models with a common fit and predict pattern.
- **PyTorch.** Flexible framework for neural networks; use it when classical models are not enough.
- **Honest evaluation.** One accuracy number on a tiny sample proves little. Use more data and the right metric for the business.

Tools and resources (official sites, not checked live)

- Pandas
- NumPy
- Scikit-learn
- PyTorch

ASSIGNMENT

Practice for lesson 4.1

Business problem: Train a purchase predictor.

Steps

1. Run the tested code.
2. Change test_size.
3. Explain why the score is not proof with only 20 rows.

Expected output: Accuracy on held-out rows, and your note on its limits.

SOLUTION (ANSWER KEY)

PYTHON

```
import pandas as pd
from sklearn.linear_model import LogisticRegression
from sklearn.model_selection import train_test_split
from sklearn.metrics import accuracy_score
df = pd.DataFrame({"visits":[1,2,8,9,3,10,2,7,1,9,8,2,6,7,3,1,9,10,2,8],
                   "bought":[0,0,1,1,0,1,0,1,0,1,1,0,1,1,0,0,1,1,0,1]})
df = df.dropna()                                    # preprocessing: remove missing rows
X_train, X_test, y_train, y_test = train_test_split(df[["visits"]], df["bought"],
test_size=0.25, random_state=0)
model = LogisticRegression().fit(X_train, y_train)
acc = accuracy_score(y_test, model.predict(X_test))
print("accuracy on held-out rows:", acc)
assert acc >= 0.8
print("ml test passed")
```

The tiny sample makes the high score unreliable. Real projects need far more rows and cross-validation.

Check yourself

1. Why hold out test rows?
   Answer: To measure on data the model has not seen.
2. Why is 20 rows too few?
   Answer: The result is unstable and may not generalise.

Lesson 4.1 of the **AI and Automation** lessons. See this lesson and its official links on skillnavora.in · All chapters and lessons · Open the full lesson guide. Tools, prices and plans change; confirm on each official site.